

SingleRAN

Base Station Supporting Multi-operator PKI Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 01 (2025-03-10).

Technical Changes

Change Description	Parameter Change	Base Station Model
Changed the default and recommended values of the certificate validity check period. For details, see: • 4.4.3.3 Using MML Commands • 4.4.5 Reconfiguration	Modified the CERTCHKTSK.PERIOD (<i>5G gNodeB, LTE eNodeB</i>) parameter.	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Revised descriptions of the principles of digital certificate usage in UMPT+UMPT cold backup mode. For details, see [4.1.3.5 Digital Certificate Usage in UMPT+UMPT Cold Backup Mode](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
GSM	GBFD-171205	BTS Supporting Multi-operator PKI	4 Base Station Supporting Multi-operator PKI
UMTS	WRFD-171220	NodeB Supporting Multi-operator PKI	
LTE FDD	LOFD-081280	eNodeB Supporting Multi-operator PKI	
LTE TDD	TDLOFD-081206	eNodeB Supporting Multi-operator PKI	
NB-IoT	MLOFD-081282	eNodeB Supporting Multi-operator PKI	
NR	FBFD-010023	Security Mechanism (gNodeB Supporting Multi-operator PKI)	

3 Overview

As network deployment demands increase, operators are confronted with the following challenges if they independently deploy networks:

- Expensive spectrum licenses
- Significant network deployment costs
- High network coverage requirements
- Difficult site deployment

To cope with these challenges, more and more operators choose the network sharing solution (RAN Sharing for short), through which they can use one set of base station equipment to cover the same area. For details about network sharing, see:

- *RAN Sharing in GBSS Feature Documentation*
- *RAN Sharing in RAN Feature Documentation*
- *RAN Sharing in eRAN Feature Documentation*
- *Multi-Operator Sharing in 5G RAN Feature Documentation*

In RAN Sharing scenarios, if a base station can only be deployed with the public key infrastructure (PKI) server of one operator (the primary operator), IPsec tunnels of secondary operators must be authenticated using the certificate issued by the PKI server of the primary operator. This means that the IPsec tunnel reliability of secondary operators depends on the PKI server of the primary operator.

To address this issue, the Base Station Supporting Multi-operator PKI feature is introduced, enabling a base station to be deployed with the PKI systems of multiple operators.

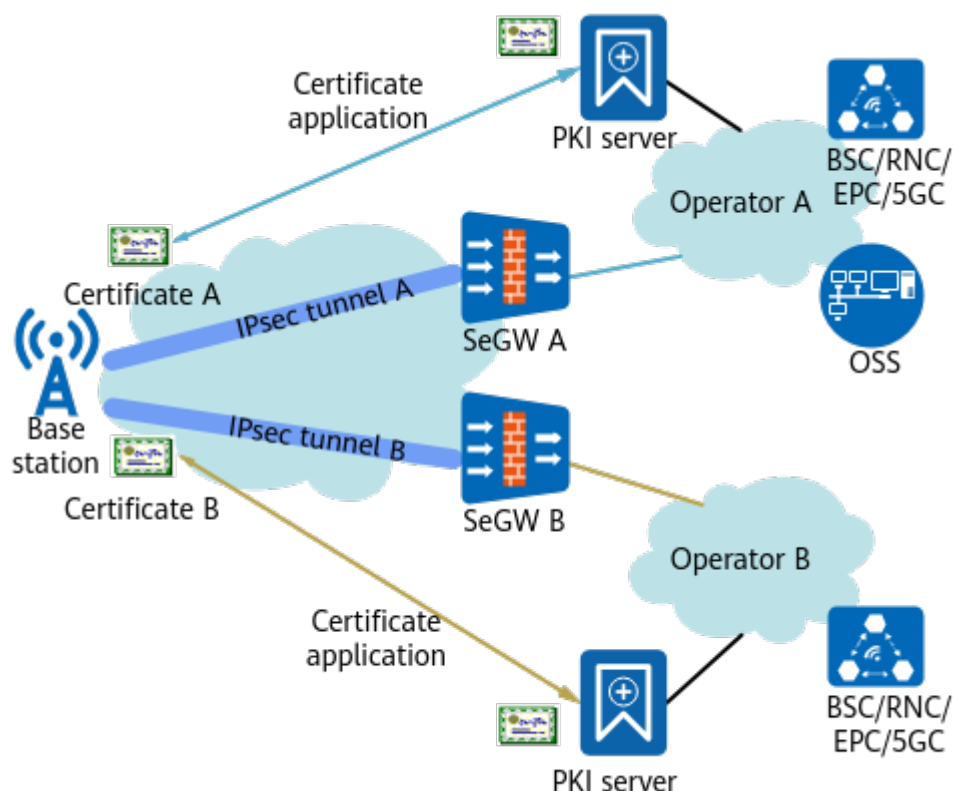
4 Base Station Supporting Multi-operator PKI

4.1 Principles

4.1.1 Introduction

This feature enables each operator to deploy its own PKI server on the base station. With this feature, loading and management of certificates from multiple operators are allowed on the base station, and certificate application, update, and revocation of one operator are independent from those of another operator. The IPsec tunnel of each operator uses the certificates issued by its own PKI server for authentication, as shown in [Figure 4-1](#).

Figure 4-1 Networking of base station supporting multi-operator PKI



Limitations

The Base Station Supporting Multi-operator PKI feature can be deployed only in RAN Sharing scenarios, but it is not supported by the GBTS.

Specifications

- When PKI redundancy is used, each base station supports a maximum of six pairs of Certificate Authorities (CAs). When PKI redundancy is not used, each base station supports a maximum of six CAs.
- Each base station supports six periodic certificate revocation list (CRL) acquisition tasks, which are configured using the **CRLTSK** managed object (MO).
- Each base station allows loading of a maximum of 20 certificates, including a Huawei initial certificate.

If operators use multi-level certificates and the certificates take up more storage space than is available, then these certificates will be converted into the .p7b format to save storage.

4.1.2 Architecture

Figure 4-2 illustrates the PKI system architecture for the Base Station Supporting Multi-operator PKI feature when two operators share a base station.

The PKI system of operator 1 consists of CA 1, RA 1, and digital certificate & CRL database 1. The PKI system of operator 2 consists of CA 2, RA 2, and digital certificate & CRL database 2. RA is short for registration authority.

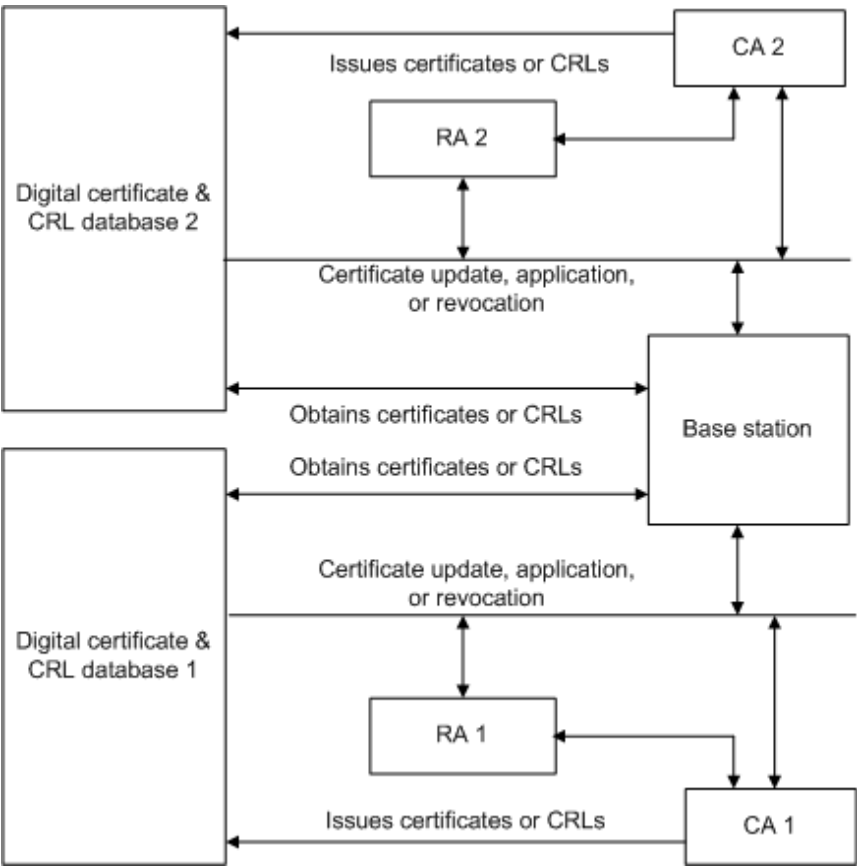
A CA is the central management node in a PKI system and is used by enterprises to issue and manage certificates.

An RA is a certificate registration and approval authority.

The digital certificate & CRL database stores all CA-issued certificates and CRLs.

For details about the CA, RA, and digital certificate & CRL database, see *PKI*.

Figure 4-2 PKI system architecture for the Base Station Supporting Multi-operator PKI feature



4.1.3 Certificate Management and Application

Table 4-1 describes the differences in certificate management and application between multi-operator PKI and single-operator PKI. For the similarities, see *PKI*.

Table 4-1 Differences between multi-operator PKI and single-operator PKI

Function		Is There a Difference?
Certificate management and application	Certificate preconfiguration phase	No
	Base station deployment phase	For the differences, see 4.1.3.2 Base Station Deployment Phase .
	Certificate application	For the differences, see 4.1.3.3.1 Certificate Application .
	Certificate sharing	No
	Certificate validity check	No
	Certificate update	No
	Certificate revocation	No
	CRL acquisition	No
	PKI networking reliability	No
	Digital certificate usage in UMPT+UMPT cold backup mode	No

4.1.3.1 Certificate Preconfiguration Phase

A base station is preconfigured with a Huawei initial certificate before delivery. When the base station supports PKIs for multiple operators, the base station uses the preconfigured initial certificate to apply for certificates from each operator.

4.1.3.2 Base Station Deployment Phase

[Figure 4-3](#) shows an IPsec networking where digital certificates are used for identity authentication.

In RAN Sharing scenarios, the base station sets up an OM channel only with the primary operator and the primary operator manages the base station. In the following figure, the PKI system of the primary operator comprises of CA 1, RA 1, and CRL server 1; the PKI system of the secondary operator comprises of CA 2, RA 2, and CRL server 2. The OM channel is protected by Secure Sockets Layer (SSL).

Figure 4-3 Networking for deploying Base Station Supporting Multi-operator PKI in RAN Sharing scenarios

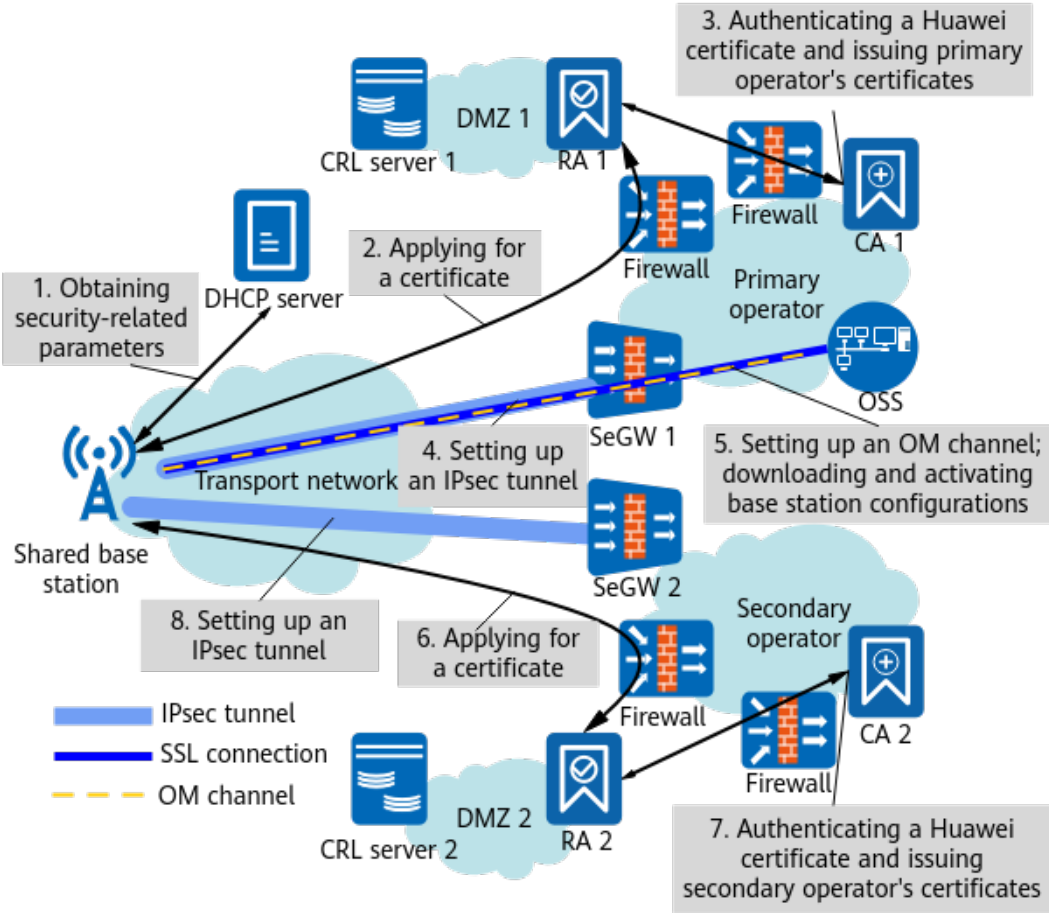
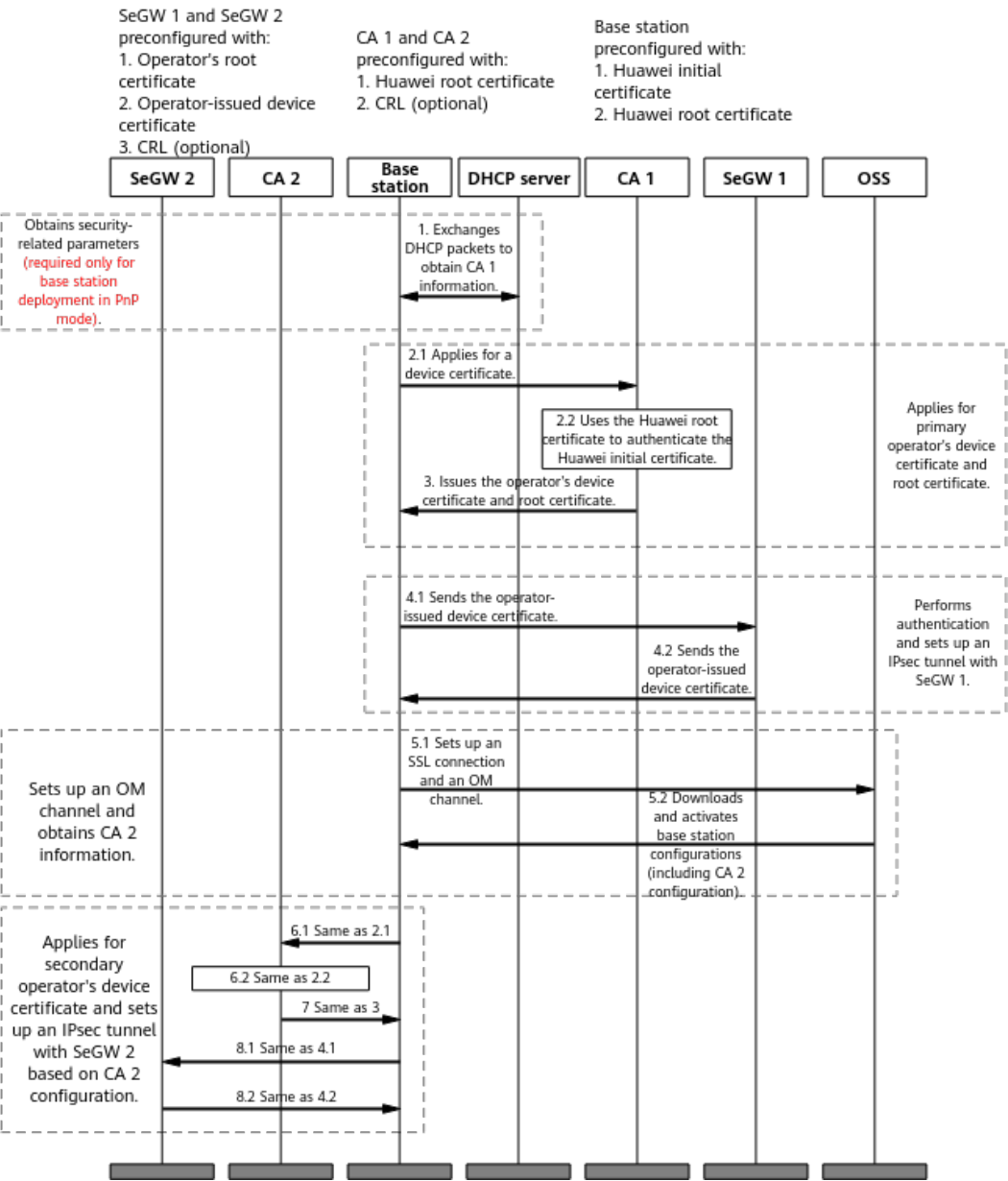


Figure 4-4 details base station deployment procedures illustrated in **Figure 4-3**.

Figure 4-4 Automatic base station deployment



NOTE

During CMPv2-based automatic certificate application, the base stations at both ends of an SSL connection use Huawei initial certificates for authentication.

Deployment of Base Station Supporting Multi-operator PKI differs from single-operator PKI deployment in the following aspects:

- Each operator's CA should be preconfigured with Huawei's root certificate and a Huawei CRL (optional), which are used to verify a Huawei initial certificate.
- Each operator's security gateway (SeGW) should be preconfigured with its own operator's root certificate, an operator's CRL (optional), and an operator-issued device certificate, which are used for the bidirectional authentication between the SeGW and the Huawei base station.

- During automatic base station deployment, the base station needs to apply for a certificate from the CAs of the two operators, and perform a bidirectional authentication with each operator's SeGW. In base station deployment by plug and play (PnP), the base station must first apply for a certificate from the CA system of the primary operator and then from the CA system of the secondary operator.

Table 4-2 illustrates the differences in MOs used for configuring multi-operator PKI compared with those used for configuring single-operator PKI.

Table 4-2 Differences in MOs

Object	Difference
CA	- The Certificate Request Switch parameter is added. - All parameters in the CERTREQ MO are added.
CERTMK	- The CA Switch parameter is added. - The Certificate Authority Name parameter is added.
IKEPEER	- The Certificate Source parameter is added. - The Certificate File Name parameter is added.

4.1.3.3 Operation Phase

The following certificate management activities are performed in the operation phase: certificate application, certificate sharing, certificate validity check, certificate update, certificate revocation, and CRL acquisition.

4.1.3.3.1 Certificate Application

Multi-operator PKI has the following requirements in the certificate application phase:

- If operators use different certificate request templates, these certificate request templates must be configured before certificate application.
Set the **CA.CERTREQSW (5G gNodeB, LTE eNodeB)** parameter to **USERDEFINE** to customize a certificate request template for the CA.
- When a manual CMPv2-based certificate application is triggered:
 - Operators' certificates must be applied for one by one. That is, the certificates of all operators cannot be applied for at a time.
 - When the **REQ DEVCERT** command is executed to trigger a CMPv2-based certificate application, the initial certificate is used for certificate application by default. There is no need to run the **MOD APPCERT** command to change a configured certificate to the initial certificate.

NOTE

After the base station sends a CMPv2-based certificate request message to the CA, the certificate application procedure fails if the certificate request times out. The waiting timeout interval is 60s in single-operator PKI scenarios and is 20s for each PKI in multi-operator PKI scenarios.

- After a successful certificate application, the obtained operator's certificate will be automatically loaded to the **CERTMK** MO, and the **CERTMK.CASW (5G gNodeB, LTE eNodeB)** parameter is automatically set to **ON** for this certificate.
- Before a reconstruction from single-operator PKI to multi-operator PKI, the **CERTMK.CASW (5G gNodeB, LTE eNodeB)** parameter must be set to **ON**.
- After a successful certificate application, the **MOD APPCERT** command can be executed to set the certificate specified by only one **CERTMK** MO as the certificate used in the **APPCERT** MO.
- After successful certificate loading, each IPsec tunnel needs to be bound to an operator's certificate.

You can use the **IKEPEER.CERTSOURCE (5G gNodeB, LTE eNodeB)** and **IKEPEER.CERTNAME** parameters to bind operators' certificates to IPsec tunnels.

NOTE

Managing multiple trust certificates through a trust group helps improve security in scenarios where multiple operators' certificates are deployed. For details about trust groups, see *PKI*.

4.1.3.3.2 Certificate Sharing

The SSL certificate sharing method in multi-operator PKI scenarios is the same as that in single-operator PKI scenarios. Secondary operators have no SSL connection and therefore, they do not need to use the SSL certificate.

4.1.3.3.3 Certificate Validity Check

In multi-operator PKI scenarios, the periodic certificate validity check task is globally set for all operators. You cannot set a periodic certificate validity check task for a specific operator.

4.1.3.3.4 Certificate Update

In multi-operator PKI scenarios, a manual CMPv2-based certificate update procedure can only be triggered for operators one by one. The automatic CMPv2-based certificate update procedure in multi-operator PKI scenarios is the same as that in single-operator PKI scenarios.

4.1.3.3.5 Certificate Revocation

The certificate revocation procedure in multi-operator PKI scenarios is the same as that in single-operator PKI scenarios.

4.1.3.3.6 CRL Acquisition

In multiple-operator PKI scenarios:

- Operators' CRL servers are independent of each other and the CRL acquisition procedure is the same as that in single-operator PKI scenarios.
- Only one global CRL policy can be configured for a base station. The global CRL policy is configured using the **CRLPOLICY** MO.
- Each base station can be configured with six periodic CRL acquisition tasks, which can be configured using the **CRLTSK** MO.

4.1.3.4 PKI Networking Reliability

To improve the reliability of PKI-based secure networks, the base station supports PKI redundancy in multi-operator PKI scenarios.

- The working mechanism of PKI redundancy in multi-operator PKI scenarios is the same as that in single-operator PKI scenarios.
- The active and standby PKI servers must belong to the same operator.
- The base station supports a maximum of six pairs of PKI servers in redundancy mode.

4.1.3.5 Digital Certificate Usage in UMPT+UMPT Cold Backup Mode

The certificate usage in UMPT+UMPT cold backup mode in multi-operator PKI scenarios is the same as the certificate usage in single-UMPT scenarios.

However, in multi-operator PKI scenarios, a base station manages the certificates of multiple operators. That is, the number of certificates managed by one base station increases. A base station can manage a maximum of 20 certificates, including a Huawei initial certificate.

4.2 Network Analysis

4.2.1 Benefits

In RAN Sharing scenarios, if each operator deploys its own PKI server, this feature provides an independent IPsec tunnel for each operator to achieve the secure isolation of each operator's services.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.1.3 Function Impacts](#), [4.3.2.2.3 Function Impacts](#), [4.3.2.3.3 Function Impacts](#), [4.3.2.4.3 Function Impacts](#), [4.3.2.5.3 Function Impacts](#), and [4.3.2.6.3 Function Impacts](#).

The duration of base station deployment is extended by about 10s each time an operator-issued certificate is applied for.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 GBFD-171205 BTS Supporting Multi-operator PKI

4.3.2.1.1 Prerequisite Functions

RA T	Function Name	Function Switch	Reference
GS M	Abis over IP	None	<i>IPv4 Transmission</i>

4.3.2.1.2 Mutually Exclusive Functions

None

4.3.2.1.3 Function Impacts

None

4.3.2.2 WRFD-171220 NodeB Supporting Multi-operator PKI

4.3.2.2.1 Prerequisite Functions

RA T	Function Name	Function Switch	Reference
UM TS	IP Transmission Introduction on lub Interface	None	<i>IPv4 Transmission</i>

4.3.2.2.2 Mutually Exclusive Functions

None

4.3.2.2.3 Function Impacts

None

4.3.2.3 LOFD-081280 eNodeB Supporting Multi-operator PKI

4.3.2.3.1 Prerequisite Functions

None

4.3.2.3.2 Mutually Exclusive Functions

None

4.3.2.3.3 Function Impacts

None

4.3.2.4 TDLOFD-081206 eNodeB Supporting Multi-operator PKI

4.3.2.4.1 Prerequisite Functions

None

4.3.2.4.2 Mutually Exclusive Functions

None

4.3.2.4.3 Function Impacts

None

4.3.2.5 MLOFD-081282 eNodeB Supporting Multi-operator PKI

4.3.2.5.1 Prerequisite Functions

None

4.3.2.5.2 Mutually Exclusive Functions

None

4.3.2.5.3 Function Impacts

None

4.3.2.6 FBFD-010023 Security Mechanism (gNodeB Supporting Multi-operator PKI)

4.3.2.6.1 Prerequisite Functions

None

4.3.2.6.2 Mutually Exclusive Functions

None

4.3.2.6.3 Function Impacts

None

4.3.3 Hardware

Base Station Models

RAT	Base Station Model
GSM	3900 and 5900 series base stations (macro base stations)
UMTS	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
LTE	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
NR	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910. - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Macro base stations: This feature is not supported by the GBTS.

Boards

NE Type	Board Configuration	Board That Provides a Port for Connecting to the Transport Network	Port Type
eGBTS	UMPT/UMDU/MDUC	UMPT/UMDU/MDUC	Ethernet port
eGBTS	UMPT+UTRPc	UTRPc	Ethernet port
NodeB	UMPT/UMDU/MDUC	UMPT/UMDU/MDUC	Ethernet port
NodeB	UMPT+UTRPc	UTRPc	Ethernet port
eNodeB	UMPT/UMDU	UMPT/UMDU	Ethernet port
eNodeB	UMPT+UTRPc	UTRPc	Ethernet port
gNodeB	UMPT	UMPT	Ethernet port

RF Modules

This function does not depend on RF modules.

4.3.4 Others

Before deploying this feature, engineering personnel must obtain CA information from CA maintenance personnel. The required CA information in this scenario is the same as that in single-PKI scenarios. For details, see *PKI*.

- The PKI server (CA) of each operator must be deployed. Each base station supports a maximum of six operators' PKI servers, that is, six independent CAs or twelve active/standby CAs.
- The device certificate and CRL file issued by each operator's CA server must meet the RFC 5280 standards.
- The operator's CA server complies with the CMPv2 specified in the RFC 4210 standards. The certificate request message format meets the RFC 4211 standards.
- The operator's CA server meets the following specification in 3GPP TS 33.310: The certificate request message contains the operator's root certificate or certificate chain.
- The operator's CA server is preconfigured with the Huawei root certificate.

4.4 Operation and Maintenance

4.4.1 When to Use

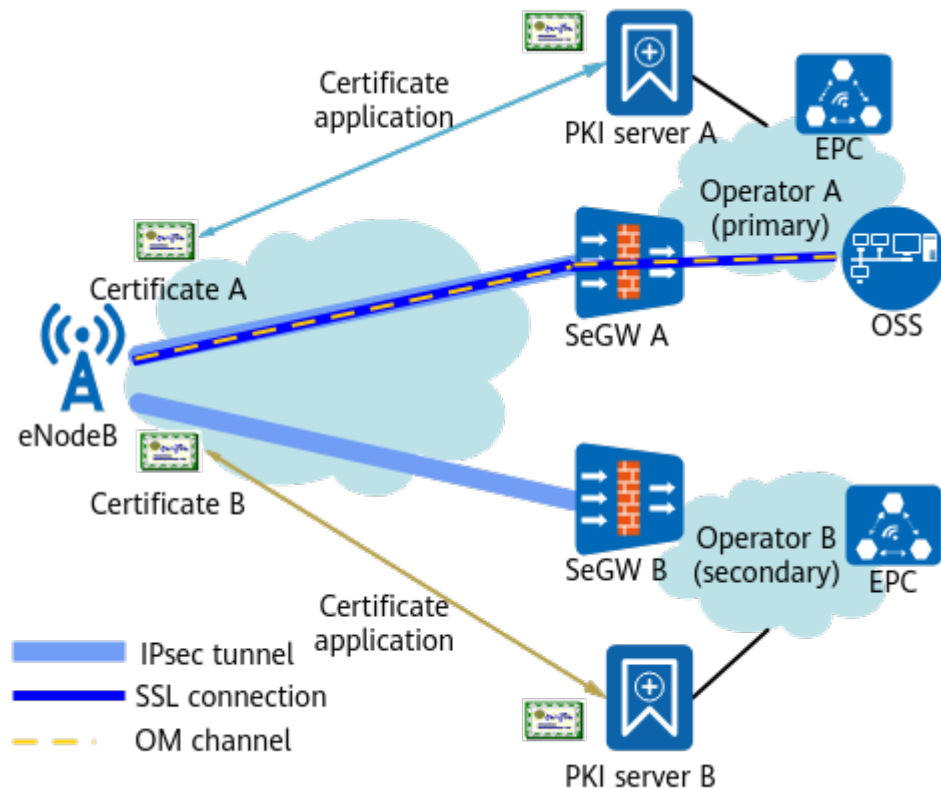
To isolate each operator's services in RAN Sharing scenarios, if each operator deploys its own PKI server, this feature must be enabled. Before feature deployment, configure PKI information for each operator.

4.4.1.1 Typical Scenarios

Single-Mode Base Station

Figure 4-5 uses an LTE-only base station as an example to illustrate the PKI system in a RAN Sharing scenario where operator A and operator B share the base station managed by operator A and the two operators have their own PKI systems.

Figure 4-5 LTE-only base station



Co-MPT Multimode Base Station

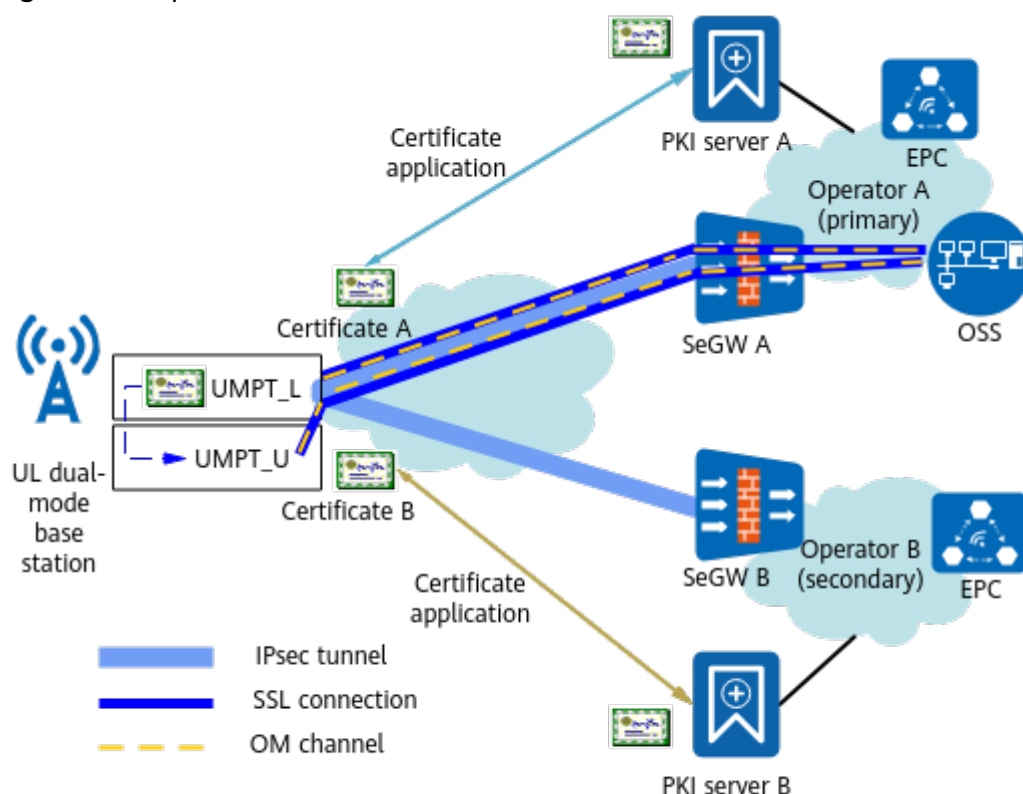
The PKI system of a co-MPT multimode base station is the same as that of a single-mode base station, as shown in [Figure 4-5](#).

Separate-MPT Multimode Base Station

[Figure 4-6](#) uses a separate-MPT UL dual-mode base station as an example. In this scenario:

- The UMPT_L and UMPT_U are shared by operator A (the primary operator) and operator B.
- UMTS data is transmitted through LTE.
- On the MAE of the primary operator, the base station is managed as two separated base stations.
- The UMPT_U and UMPT_L have a separate SSL connection and OM channel with the MAE. The UMPT_U shares the SSL certificate with the UMPT_L.
- The UMPT_L has a separate IPsec tunnel with both SeGW A and SeGW B. The two IPsec tunnels are authenticated using the certificate issued by the corresponding operator.
- The two operators' certificates are deployed on the UMPT_L.

Figure 4-6 Separate-MPT UL base station

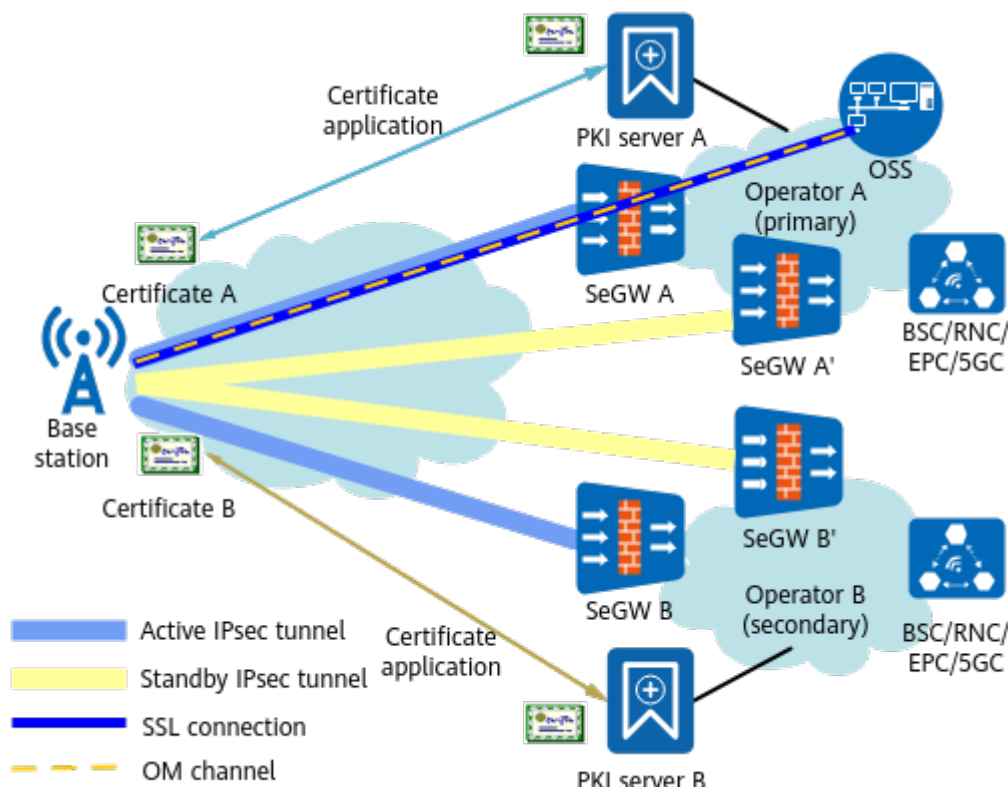


IPsec Redundancy Among Multiple SeGWs

IPsec redundancy among multiple SeGWs improves the reliability of base station operation. As shown in [Figure 4-7](#), to support the Base Station Supporting Multi-operator PKI feature and enable secure isolation of different operator networks, two SeGWs corresponding to the primary and standby IPsec tunnels are recommended for each operator.

Before deploying the Base Station Supporting Multi-operator PKI feature, enable IPsec redundancy among multiple SeGWs. For details, see *IPsec*. For details about how to configure the Base Station Supporting Multi-operator PKI feature in IPsec redundancy mode, see [4.4.3 Data Configuration](#).

Figure 4-7 Multi-operator PKI enabled with IPsec redundancy among multiple SeGWs



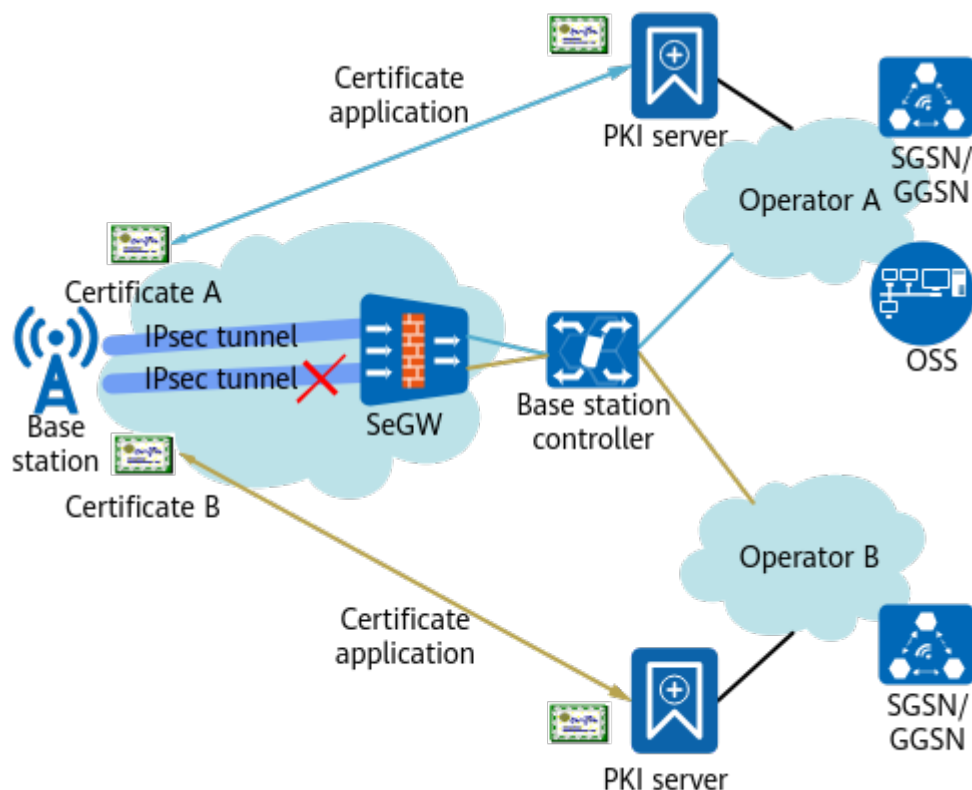
4.4.1.2 Unrecommended Scenario

Shared Base Station Controller with No IPsec Tunnel Between the Base Station Controller and CN (GSM/UMTS)

Two operators share the base station controller, which is connected to the CN of each operator, and no IPsec tunnel is set up between the base station controller and the CNs, as shown in [Figure 4-8](#).

In this scenario, if different IPsec tunnels are set up between the base station and base station controller for data isolation for different operators, data of the two operators is still converged on the base station controller and then forwarded to their respective CN. Therefore, it is recommended that only one IPsec tunnel be set up between the base station and base station controller, and the primary operator's digital certificate and SeGW be used.

Figure 4-8 Shared base station controller with no IPsec tunnel between the base station controller and CN

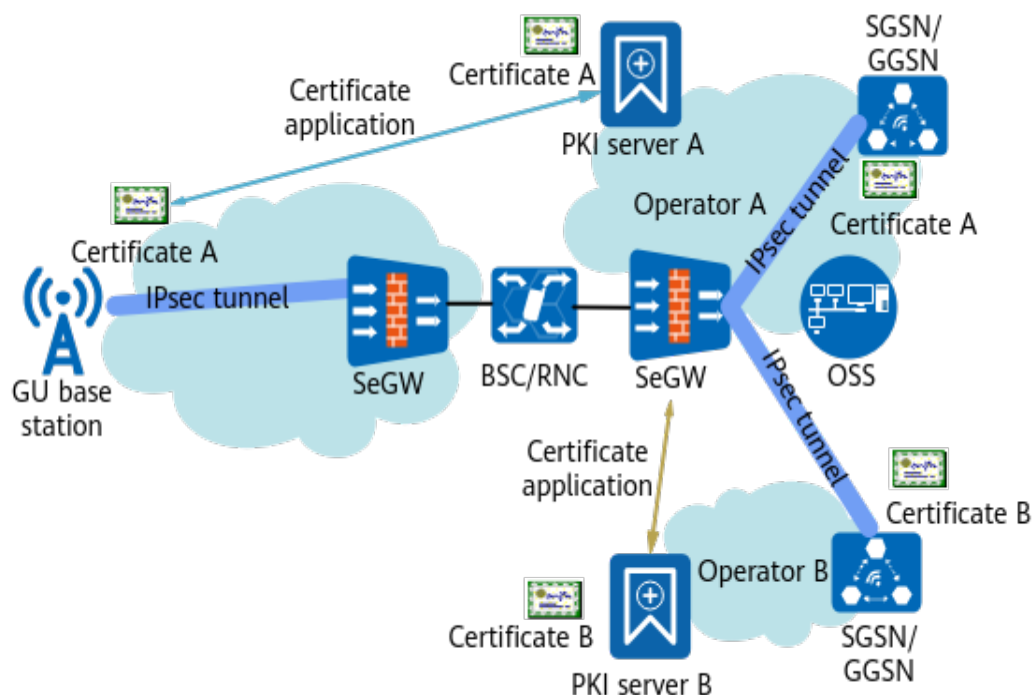


Shared Base Station Controller with IPsec Tunnels Between the Base Station Controller and CNs (GSM/UMTS)

Two operators share the base station controller, which is connected to the CN of each operator, and IPsec tunnels are set up between the base station controller and the CNs, as shown in [Figure 4-9](#).

In this scenario, although the base station controller has separate IPsec tunnels with the CNs, the base station controller supports the IPsec tunnel only with an external SeGW. If separate IPsec tunnels are to be set up between the base station and base station controller for data isolation for different operators, different digital certificates must be configured for the operators to authenticate these IPsec tunnels and certificate update should be performed separately for different PKI systems.

Figure 4-9 Shared base station controller with IPsec tunnels between the base station controller and CNs



4.4.1.3 Forbidden Scenarios

- In a GU RAN Sharing network, operators share the base station but use different base station controllers.
At present, the GU dual-mode base station cannot be connected to base station controllers of different operators.
- OM channels are securely isolated.
In RAN Sharing scenarios, the base station does not support separate OM channels for different operators and only the primary operator can set up the SSL-based OM channel. In this case, this feature cannot implement secure isolation of OM channels.

4.4.2 Precautions

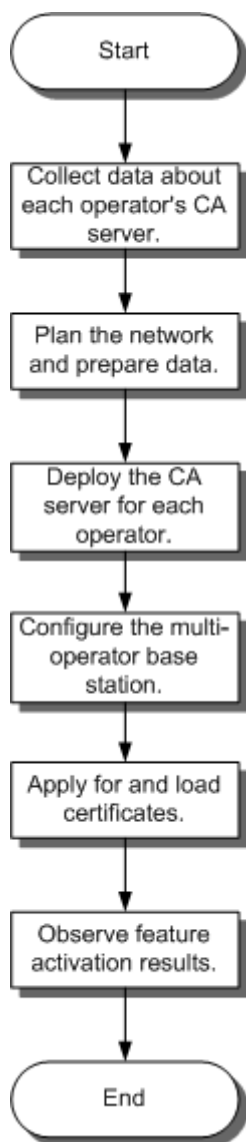
During new PKI deployment, the IPsec tunnel needs to be reestablished, which interrupts services.

4.4.3 Data Configuration

4.4.3.1 Deployment Process

Figure 4-10 shows the feature deployment process.

Figure 4-10 Process of deploying the Base Station Supporting Multi-operator PKI feature



4.4.3.2 Data Preparation

Data needs to be prepared for enabling the Base Station Supporting Multi-operator PKI feature. For parameters related to the PKI and PKI redundancy features, see *PKI*. For parameters related to IPsec Redundancy Among Multiple SeGWs, see *IPsec*.

The base station must initiate certificate application requests to the CA server of each operator. Each operator's CA information must be configured on the base station side. The involved MO is **CA**. [Table 4-3](#) describes the parameters to be configured in this MO.

Table 4-3 Data to be prepared on the base station side for the CA server

Parameter Name	Parameter ID	Setting Notes
Certificate Request Switch	CA.CERTREQSW (5G gNodeB, LTE eNodeB)	- When the certificate request template configured in the MOD CERTREQ command is used, set this parameter to DEFAULT(DEFAULT). - When a customized certificate request template is used, set this parameter to USERDEFINE(USERDEFINE).
Common Name Additional Info.	CA.USERADDINFO (5G gNodeB, LTE eNodeB)	These parameters are valid only when CERTREQSW is set to USERDEFINE(USERDEFINE). These parameters are used to configure the certificate request template used for certificate application for a secondary operator. The setting notes are the same as those in the CERTREQ MO.
Country	CA.COUNTRY (5G gNodeB, LTE eNodeB)	
Organization	CA.ORG (5G gNodeB, LTE eNodeB)	
Organizational Unit	CA.ORGUNIT (5G gNodeB, LTE eNodeB)	
State or Province	CA.STATEPROVINCE NAME (5G gNodeB, LTE eNodeB)	
Locality	CA.LOCALITY (5G gNodeB, LTE eNodeB)	
Key Usage	CA.KEYUSAGE (5G gNodeB, LTE eNodeB)	
Certificate Request Signature Algorithm	CA.CERTREQSIGNALG (5G gNodeB, LTE eNodeB)	
Key Size	CA.KEYSIZE (5G gNodeB, LTE eNodeB)	
Local Name	CA.LOCALNAME (5G gNodeB, LTE eNodeB)	

Parameter Name	Parameter ID	Setting Notes
Local IP	<i>C.A.LOCALIP (5G gNodeB, LTE eNodeB)</i>	

Table 4-4 lists the data to be prepared for a device certificate (the **CERTMK** MO).

Table 4-4 Data to be prepared for a device certificate

Parameter Name	Parameter ID	Setting Notes
CA Switch	<i>CERTMK.CASW (5G gNodeB, LTE eNodeB)</i>	- When CMPv2-based feature deployment is used, bind certificates issued for all operators to the corresponding CA. In this case, set this parameter to ON(On) for each operator's certificate. - Set this parameter to OFF(Off) for Huawei initial certificates.
Certificate Authority Name	<i>CERTMK.CANAME (5G gNodeB, LTE eNodeB)</i>	This parameter is valid only when CASW is set to ON(On) .

Table 4-5 lists the data to be prepared for an IKE peer (the **IKEPEER** MO).

Table 4-5 Data to be prepared for the IKE peer

Parameter Name	Parameter ID	Setting Notes
Certificate Source	<i>IKEPEER.CERTSOURC E (5G gNodeB, LTE eNodeB)</i>	In multi-operator PKI scenarios, you need to bind a certificate for each IKEPEER MO. - If the certificate configured by the APPCERT MO is used, set this parameter to APPCERT(Appcert). - If the certificate configured by the CERTMK MO is used, set this parameter to CERTMK(Certmk).
Certificate File Name	<i>IKEPEER.CERTNAME</i>	This parameter is valid only when CERTSOURCE is set to CERTMK(Certmk) .

4.4.3.3 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

- From no-PKI to multi-operator PKI

If the following conditions are met on a base station where PKI has not been enabled:

- Operator A: C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1
- Operator B: C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca2

```
//Specifying the board where a certificate is to be deployed and resetting the base station for the configuration to take effect (If the base station has only one main control board, the certificate is deployed on the main control board by default. In this case, skip this step.)  
SET CERTDEPLOY: DEPLOYTYPE=SPECIFIC, CN=0, SRN=0, SN=7;  
//Configuring a global certificate request template (If the certificate request file used by the CA is the same as the global certificate request template, use the template specified in this step. If they are different, configure a certificate request template for the CA in the next step.)  
MOD CERTREQ: COMMNAME=ESN, USERADDINFO=".huawei.com", COUNTRY="cn", ORG="ITEF",  
ORGUNIT="Hw", STATEPROVINCENAME="sc", LOCALITY="cd",  
KEYUSAGE=DATA_ENCIPHERMENT-1&DIGITAL_SIGNATURE-1&KEY_AGREEMENT-1&KEY_ENCIPHERMENT-1,  
SIGNALG=SHA256, KEYTYPE=RSA, KEYSIZE=KEYSIZE2048,  
LOCALNAME="abcdefghijkmn.huawei.com", LOCALIP="10.20.20.188";
```

The following two scenarios are involved for setting CA server information for operator A and using this information to customize a certificate request template for the CA.

- Scenario 1:

```
//If operator A's CA server can be accessed through either the internal or public network and the OM data is protected by IPsec, it is recommended that the source IP address used for certificate application be set to an interface IP address, the source IP address used for certificate update be set to the OM IP address (for example, 10.31.31.188), the IP address in the CA URL for site deployment be set to 10.87.87.87, and a user-defined certificate request switch setting be used. The following is an example:  
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1",  
URL="http://10.88.88.88:80/pkix/", SIGNALG=SHA256, MODE=CFG_INIT_UPD_ADDR,  
UPDSIP="10.31.31.188", INITREQURL="http://10.87.87.87:80/pkix/", INITREQSIP="10.20.20.188",  
CERTREQSW=USERDEFINE, COUNTRY="cn", ORG="ITEF", ORGUNIT="hw",  
STATEPROVINCENAME="sc", LOCALITY="cd",  
KEYUSAGE=DATA_ENCIPHERMENT-1&DIGITAL_SIGNATURE-1&KEY_AGREEMENT-1&KEY_ENCIPHERMENT-1,  
CERTREQSIGNALG=SHA256, KEYTYPE=RSA, KEYSIZE=KEYSIZE2048;
```

- Scenario 2:

```
//If operator A's CA server can be accessed through either the internal or public network and the OM data is not protected by IPsec, it is recommended that the source IP address used for certificate update be set to an internal network IP address (for example, 10.45.45.45), the source IP address used for certificate application be set to an interface IP address, the IP address in the CA URL for site deployment be set to 10.87.87.87, and the default certificate request switch setting be used.  
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1",  
URL="http://10.88.88.88:80/pkix/", SIGNALG=SHA256, MODE=CFG_INIT_UPD_ADDR,
```

```

UPDSIP="10.45.45.45", INITREQURL="http://10.87.87.87:80/pkix/", INITREQSIP="10.20.20.188",
CERTREQSW=DEFAULT;

//Using an interface IP address for certificate application and certificate update, and setting the
certificate request switch to the default configuration if operator A uses PKI redundancy
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1", URL="http://
10.88.88.88:80/pkix/", SIGNALG=SHA256, MODE=CFG_INIT_UPD_ADDR, UPDSIP="10.45.45.45",
INITREQURL="http://10.85.85.85:80/pkix/", INITREQSIP="10.20.20.188", SLVURL="http://10.10.10.87:80/
pkix/", SLVINITREQURL="http://10.10.10.86:80/pkix/", CERTREQSW=DEFAULT;
//Setting CA server information for operator B
//If operator B's CA server can be accessed through only the public network, it is recommended that
the interface IP address be used for certificate application and certificate update, and a user-defined
certificate request switch setting be used.
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca2", URL="http://
10.89.89.89:80/pkix/", SIGNALG=SHA256, MODE=CFG_INIT_UPD_ADDR, UPDSIP="10.20.20.188",
INITREQURL="http://10.86.86.86:80/pkix/", INITREQSIP="10.35.35.35", CERTREQSW=USERDEFINE,
COMMONNAME=ESN, USERADDINFO=".huawei.com", COUNTRY="cn", ORG="ITEF", ORGUNIT="hw",
STATEPROVINCENAME="sc", LOCALITY="cd",
KEYUSAGE=DATA_ENCIPHERMENT-1&DIGITAL_SIGNATURE-1&KEY_AGREEMENT-1&KEY_ENCIPHERME
NT-1, CERTREQSIGNALG=SHA256, KEYTYPE=RSA, KEYSIZE=KEYSIZE2048;
//Using an interface IP address for certificate application and certificate update, and setting the
certificate request switch to the default configuration if operator B uses PKI redundancy
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca2", URL="http://
10.89.89.89:80/pkix/", SIGNALG=SHA256, MODE=CFG_INIT_UPD_ADDR, UPDSIP="10.35.35.35",
INITREQURL="http://10.86.86.86:80/pkix/", INITREQSIP="10.20.20.188", SLVURL="http://10.10.10.85:80/
pkix/", SLVINITREQURL="http://10.10.10.84:80/pkix/", CERTREQSW=DEFAULT;
//(Manual triggering of CMPv2-based certificate application) Downloading each operator's root
certificate from the FTP server (If the FTP server is deployed on the MAE, the IP address of the FTP
server is the same as that of the MAE.)
//Downloading operator A's root certificate
DLD CERTFILE: IP="10.60.60.60", USR="admin", PWD="*****", SRCF="OperationCA1.cer",
DSTF="OperationCA1.cer";
//Downloading operator B's root certificate
DLD CERTFILE: IP="10.60.60.60", USR="admin", PWD="*****", SRCF="OperationCA2.cer",
DSTF="OperationCA2.cer";
//(Manual triggering of CMPv2-based certificate application) Setting each operator's root certificate
to the trust certificate
//Setting operator A's root certificate as the trust certificate
ADD TRUSTCERT: CERTNAME="OperationCA1.cer";
//Setting operator B's root certificate as the trust certificate
ADD TRUSTCERT: CERTNAME="OperationCA2.cer";
//(Manual triggering of CMPv2-based certificate application) Setting information used by the base
station to apply for operator-issued device certificates
//Manually applying for a digital certificate for operator A
REQ DEVCERT: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1",
APPCERT="OPKIDevCert1.cer";
//Manually applying for a digital certificate for operator B
REQ DEVCERT: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca2",
APPCERT="OPKIDevCert2.cer";
//Setting information about a global certificate (If operator A's certificate is used as the global
certificate, operators not deployed with PKI servers can share this certificate.)
MOD APPCERT: APPTYPE=IKE, APPCERT="OPKIDevCert1.cer";
//Configuring the certificate used for IKE negotiation
//Enabling operator A to use the global certificate for IKE negotiation
ADD IKEPEER: PEERNAME="peer", PROPID=1, IKEVERSION=IKE_V2, IPVERSION=IPV4, IDTYPE=FQDN,
REMOTEIP="10.90.90.90", DPD=PERIODIC, REDUNDANCYFLAG=NONE, CERTSOURCE=APPCERT,
IPSECPREFRGSW=OFF, IKEV2FRGSW=OFF;
//Enabling operator B to use a non-global certificate (OpkiDevCert2.cer) for IKE negotiation
ADD IKEPEER: PEERNAME="peer", PROPID=1, IKEVERSION=IKE_V2, IPVERSION=IPV4, IDTYPE=FQDN,
REMOTEIP="10.91.91.91", DPD=PERIODIC, REDUNDANCYFLAG=NONE, CERTSOURCE=CERTMK,
CERTNAME="OpkiDevCert2.cer", IPSECPREFRGSW=OFF, IKEV2FRGSW=OFF;
//Setting a periodic certificate validity check task universally for all operators
SET CERTCHKTSK: PERIOD=1, ALMRNG=30, UPDTEMETHOD=CMP;
//(Optional) Manually downloading the CRL file from the FTP server (If the FTP server is deployed on
the MAE, the IP address of the FTP server is the same as that of the MAE.)
DLD CERTFILE: IP="10.60.60.60", USR="admin", PWD="*****", SRCF="eNodeB.crl", DSTF="eNodeB.crl";
//(Optional) Loading the manually downloaded CRL file
//Loading the CRL file for operator A
ADD CRL: CERTNAME="eNodeB1.crl";
//Loading the CRL file for operator B

```

```
ADD CRL: CERTNAME="eNodeB2.crl";
//(Optional) Setting the CRL policy universally for all operators
SET CRLPOLICY: CRLPOLICY= NOVERIFY;
//(Optional) Adding a periodic CRL download task
//Adding a periodic CRL download task for operator A
ADD CRLTSK: IP="10.86.86.86", USR="admin", PWD="*****", FILENAME="eNodeB1.crl",
ISCRLTIME=DISABLE, PERIOD=24, TSKID=0, CRLGETMETHOD=FTP;
//Adding a periodic CRL download task for operator B
ADD CRLTSK: IP="10.87.87.87", USR="admin", PWD="*****", FILENAME="eNodeB2.crl",
ISCRLTIME=DISABLE, PERIOD=24, TSKID=0, CRLGETMETHOD=FTP;
//Manually triggering a certificate update
//Manually updating operator A's certificate
UPD DEVCERT: APPCERT="OPKIDevCert1.cer",REKEY=YES;
//Manually updating operator B's certificate
UPD DEVCERT: APPCERT="OPKIDevCert2.cer",REKEY=YES;
```

NOTE

- After the **MOD APPCERT** command is executed, if the IKE connection is authenticated using a certificate and the status of the IKE SA is normal, the base station automatically triggers IKE renegotiation.
- If IKE or SSL negotiation is in progress on the base station when the **UPD DEVCERT** command is executed to update a certificate, the certificate update needs to be performed after the negotiation is completed.
- From single-operator PKI to multi-operator PKI
If the following conditions are met on a base station where PKI, PKI redundancy, or IPsec redundancy among multiple SeGWs has been enabled:
 - Operator A is the primary operator and operator B is a secondary operator. Before the reconstruction, the two operators use the certificate issued by operator A's PKI server for authentication. After the reconstruction, operator B uses an independent PKI server.
 - Operator A: C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1
 - Operator B: C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca2

```
//Specifying a CA for the primary operator's certificate loaded on the base station, querying the base
station device certificate, and turning on the CA switch in the CERTMK MO
LST CERTFILE: CN=0, SRN=0, SN=7, CERTFILETYPE=DEVCERT;
MOD CERTMK: APPCERT="opki1.cer", CASW=ON, CANAME="C = AU, S = Some-State, O = Internet
Widgits Pty Ltd, CN = eca1";
//Adding operator B's CA server information and customizing a certificate request template for the CA
server (If operator B's CA server can be accessed only from the public network, it is recommended
that the interface IP address be used for certificate application and update and a customized
certificate request template be used.)
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca2", URL="http://
10.89.89.89:80/pkix/", SIGNALG=SHA256, MODE=CFG_INIT_UPD_ADDR, UPDSIP="10.20.20.188",
INITREQURL="http://10.86.86.86:80/pkix/", INITREQSIP="10.35.35.35", CERTREQSW=USERDEFINE,
COMMNAME=ESN, USERADDINFO=".huawei.com", COUNTRY="cn", ORG="ITEF", ORGUNIT="hw",
STATEPROVINCENAME="sc", LOCALITY="cd",
KEYUSAGE=DATA_ENCIPHERMENT-1&DIGITAL_SIGNATURE-1&KEY_AGREEMENT-1&KEY_ENCIPHERME
NT-1, CERTREQSIGNALG=SHA256, KEYTYPE=RSA, KEYSIZE=KEYSIZE2048;
//(Manual triggering of CMPv2-based certificate application) Downloading operator B's root
certificate from the FTP server (If the FTP server is deployed on the MAE, the IP address of the FTP
server is the same as that of the MAE.)
DLD CERTFILE: IP="10.60.60.60", USR="admin", PWD="*****", SRCF="OperationCA2.cer",
DSTF="OperationCA2.cer";
//(Manual triggering of CMPv2-based certificate application) Setting operator B's root certificate as
the trust certificate (If multi-level CAs are deployed in an operator's PKI system, a complete certificate
chain must be added. If the certificates of different levels of CAs are stored separately in the
certificate chain, run the following command for each certificate to be added.)
ADD TRUSTCERT: CERTNAME="OperationCA2.cer";
//(Manual triggering of CMPv2-based certificate application) Applying for operator B's device
certificate
```

```

REQ DEVCERT: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca2",
APPCERT="OPKIDevCert2.cer";
//Configuring the certificate used for IKE negotiation (assuming that the base station has been
configured with IKE peers, a customized certificate is used for IKE negotiation for operator B, and the
certificate name is OpkiDevCert2.cer)
MOD IKEPEER: PEERNAME="peer", PROPID=1, IKEVERSION=IKE_V2, IPVERSION=IPV4, IDTYPE=FQDN,
REMOTEIP="10.91.91.91",DPD=PERIODIC, REDUNDANCYFLAG=NONE, CERTSOURCE=CERTMK,
CERTNAME="OpkiDevCert2.cer";
//(Optional) Downloading the CRL file from the FTP server (If the FTP server is deployed on the MAE,
the IP address of the FTP server is the same as that of the MAE.)
DLD CERTFILE: IP="10.60.60.60", USR="admin", PWD="*****", SRCF="eNodeB.crl", DSTF="eNodeB.crl";
//(Optional) Loading the CRL file for operator B
ADD CRL: CERTNAME="eNodeB2.crl";
//(Optional) Adding a periodic CRL download task for operator B
ADD CRLTSK: IP="10.87.87.87", USR="admin", PWD="*****", FILENAME="eNodeB2.crl",
ISCRLLTIME=DISABLE, PERIOD=24, TSKID=0, CRLGETMETHOD=FTP;

```

Optimization Command Examples

N/A

Deactivation Command Examples

- From multi-operator PKI to no-PKI

```

//Removing the binding relationships between IPsec policy groups and ports
RMV IPSECBIND: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=ETH,PN=0, SPGN="A";
RMV IPSECBIND: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=ETH,PN=0, SPGN="A";
//Removing IPsec policies
RMV IPSECPOLICY: SPGN="A", SPSN=10;
RMV IPSECPOLICY: SPGN="B", SPSN=11;
//Removing IKE peers
RMV IKEPEER: PEERNAME="ike1";
RMV IKEPEER: PEERNAME="ike2";
//Restoring a Huawei initial certificate (Skip this step if no operator-issued certificate is bound.)
MOD APPCERT: APPTYPE=IKE, APPCERT="appcert.pem";
//Removing the certificates loaded to the base station
RMV CERTMK: APPCERT="eNodeBCert1.pem";
RMV CERTMK: APPCERT="eNodeBCert2.pem";
//Removing the CAs configured for the base station
RMV CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1";
RMV CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca2";
//Removing the periodic CRL acquisition task started for multiple operators
RMV CRLTSK: TSKID=0;
RMV CRLTSK: TSKID=1;

```

NOTE

There are reference relationships between the **IKEPEER** MO and **CERTMK** MO and between the **CERTMK** MO and **CA** MO. Before running the **RMV CERTMK** and **RMV CA** commands, remove the reference relationships between these MOs.

- From multi-operator PKI to single-operator PKI

```

//(Optional, applicable only when the IKE certificate under the APPCERT MO is not the primary
operator's certificate) Changing the IKE certificate under the APPCERT MO to the primary operator's
certificate
MOD APPCERT: APPTYPE=IKE, APPCERT="eNodeBCert1.pem";
//Modifying the binding relationships between operator B's IKE and the certificate (Certificate Source
= APPCERT, which means that operator B shares the certificate with operator A; the IKE peer name for
operator B is assumed to be ike2)
MOD IKEPEER: PEERNAME="ike2", CERTSOURCE=APPCERT;
//Removing the secondary operator's certificate loaded to the base station (Assume that the
certificate file name is eNodeBCert2.pem.)
RMV CERTMK: APPCERT="eNodeBCert2.pem";
//Removing a secondary operator's CA configured for the base station
RMV CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca2";
//Changing the value of CA Switch to OFF for the primary operator's certificate that will still be used
MOD CERTMK: APPCERT="eNodeBCert1.pem", CASW=OFF;
//Changing the value of Certificate Request Switch to DEFAULT for the primary operator's certificate

```

```
MOD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1", URL="http://10.88.88.88:80/pkix/", CERTREQSW=DEFAULT;  
//Removing the periodic CRL acquisition task started for secondary operators (Assume that the task ID is 1.)  
RMV CRLTSK: TSKID=1;
```

NOTE

There are reference relationships between the **IKEPEER** MO and **CERTMK** MO and between the **CERTMK** MO and **CA** MO. Before running the **RMV CERTMK** and **RMV CA** commands, remove the reference relationships between these MOs.

4.4.3.4 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.4 Activation Verification

Step 1 Run the **DSP APPCERT** command to query the status of the global device certificate.

If the values of **Certificate File Name**, **Issuer**, and **Common Name** are correct and the value of **Status** is **Normal**, the device certificate has been loaded to the base station.

Step 2 Run the **DSP CERTMK** command to query the binding relationships between a certificate and the CA.

If the value of **CA Switch** in the returned result is **ON**, this feature has been enabled. You can query the value of **CA** to check the CA server that issues the certificate.

Step 3 Run the **DSP IKEPEER** command to query the certificate used for IKE negotiation.

Check whether the certificate has taken effect by querying the values of **Certificate Source** and **Certificate File Name**.

Step 4 Run the **DSP TRUSTCERT** command to query the status of the trust certificate.

If the value of **Status** is **Normal** in the query result, the trust certificate has been loaded to the base station.

Step 5 (Optional) Run the **DSP CRL** command to query the status of the CRL file.

If the value of **Status** is **Normal** in the query result, the CRL has been loaded to the base station.

----End

4.4.5 Reconfiguration

Reconfiguration of CA Name

In **CA.CANAME**, the S and ST fields are regarded as the same field. Services can be properly provided regardless of whether the field name is **S** or **ST**.

To change the field name from **S** to **ST**, perform the following steps:

- Step 1** Run the **ADD CA** command to add a CA.
- Step 2** Run the **MOD CERTMK** command to modify the device certificate.
- Step 3** Run the **RMV CA** command to remove the old CA.

----End

MML command examples are as follows:

```
ADD CA: CANAME="C = AU, ST = Some-State, O = Internet Widgits Pty Ltd, CN = eca1", URL="http://10.89.89.89:80/pkix/", SIGNALG=SHA256, MODE=CFG_INIT_UPD_ADDR, UPDSIP="10.20.20.188", INITREQURL="10.86.86.86:80/pkix/", INITREQSIP="10.35.35.35", CERTREQSW=USERDEFINE, COMMNAME=ESN, USERADDINFO=".huawei.com", COUNTRY="cn", ORG="ITEF", ORGUNIT="hw", STATEPROVINCENAME="sc", LOCALITY="cd", KEYUSAGE=DATA_ENCIPHERMENT-1&DIGITAL_SIGNATURE-1&KEY_AGREEMENT-1&KEY_ENCIPHERMENT-1, CERTREQSIGNALG=SHA256, KEYTYPE=RSA, KEYSIZE=KEYSIZE2048;  
MOD CERTMK: APPCERT="opki1.cer", CASW=ON, CANAME="C = AU, ST = Some-State, O = Internet Widgits Pty Ltd, CN = eca1";  
RMV CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1";
```

Certificate Reconfiguration Pre-determination

If the following commands are involved in certificate reconfiguration, the system estimates whether services will be affected after the reconfiguration. For details, see the "Estimation of Certificate Reconfiguration Impact" section in *PKI*.

- **MOD CERTREQ**
- **ADD CA**
- **MOD CA**
- **MOD APPCERT**
- **MOD CERTMK**

Activating Automatic Certificate Application After a CA Change (in Base Station Deployment/IKE Negotiation Failure Scenarios)

If the **CA.CANAME (LTE eNodeB, 5G gNodeB)** parameter is set to the RA name in a CA record, remove this CA record and then reconfigure a correct one by performing the following steps:

- Run the **ADD CA** command to add a correct CA.
- Run the **MOD CERTMK** command to change the CA to the new CA.
- Run the **RMV CA** command to remove the old CA.
- Run the **SET CERTCHKTSK** command to turn on the automatic application switch.

MML command examples are as follows:

```
//Assume that the expected RANAME is as follows: C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca2, CANAME is C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1
```

```
//The following record exists.
```

```
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca2", URL="http://10.88.88.88:80/pkix/", SIGNALG=SHA256, MODE=
```

CFG_INIT_UPD_ADDR, UPDSIP="10.31.31.188",INITREQURL="http://10.89.89.89:80/pkix/",INITREQSIP="10.20.20.188"; //

Run the following commands:

```
ADD CA:CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1",RANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca2", URL="http://10.88.88.88:80/pkix/",  
SIGNALG=SHA256, MODE= CFG_INIT_UPD_ADDR, UPDSIP="10.31.31.188",INITREQURL="http://  
10.89.89.89:80/pkix/",INITREQSIP="10.20.20.188";  
MOD CERTMK:APPCERT="opki1.cer",CASW=ON,CANAME="C = AU, S = Some-State, O = Internet Widgits  
Pty Ltd, CN = eca1";  
RMV CA: CANAME="C=AU, ST=Some-State, O=Internet Widgits Pty Ltd, CN=eca2";  
SET CERTCHKTSK: PERIOD=1, ALMRNG=30, UPDATEMETHOD=CMP,AUTOREAPPLYSW = ON;
```

4.4.6 Network Monitoring

None

5 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

- Step 2** On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.
- End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 33.310, "Network Domain Security (NDS); Authentication Framework (AF)"
- IETF RFC 4210, "Internet X.509 Public Key Infrastructure Certificate Management Protocol (CMP)"
- IETF RFC 4211, "Internet X.509 Public Key Infrastructure Certificate Request Message Format (CRMF)"
- IETF RFC 5280, "Internet X.509 Public Key Infrastructure Certificate and CRL Profile"
- IETF RFC 2585, "Internet X.509 Public Key Infrastructure Operational Protocols: FTP and HTTP"
- *IPsec*
- *PKI*
- *SRAN Networking and Evolution Overview*
- *IPv4 Transmission*
- *Multi-Operator Sharing*
- *S1 and X2 Self-Management* in eRAN feature documentation
- *NG and Xn Self-Management* in 5G RAN feature documentation
- *License Control Item Lists (FDD), License Control Item Lists (CIoT), and License Control Item Lists (TDD)* in eRAN feature documentation
- *License Control Item Lists* in 5G RAN feature documentation